

Computer Programming with MATLAB



Lesson 3: Functions

by

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Formal Definition

▶ **function** [out_arg1, out_arg2, ...] =
 function_name (in_arg1, in_arg2, ...)

▶ Examples

- **function** func
- **function** func(in1)
- **function** func(in1, in2)
- **function** out1 = func
- **function** out1 = func(in1)
- **function** [out1, out2] = func
- **function** [out1, out2] = func(in1, in2)
- ...

Function names

- ▶ Use meaningful names that tell you something about what your function does
- ▶ Do not use existing names, e.g., plot, sum, sqrt, sin, etc.
 - MATLAB already has these
 - It would get really confusing really fast
 - To check whether a name is already in use, try the built-in exist function. To see how it works, try

```
>> help exist
```

Advantages of functions

- ▶ Functions allow you to break down large, complex problems to smaller, more manageable pieces
- ▶ Functional decomposition
- ▶ Reusability
- ▶ Generality
 - A function can solve a set of related problems not just a specific one by accepting input arguments.
 - For example, the built-in function **plot** can draw a wide range of figures based on its input